# STACKING WIZARD

&

# ASTRO WIZARD



# USER MANUAL

Professional Astrophotography –
easy and uncomplicated.
Stack. Optimize. Enhance.

**AUTOMATE**
Simplify and automate processes

**STACK**
Intelligent stacking for maximum detail

**OPTIMIZE**
Improve and perfect your captures

**ENHANCE**
Make results brilliant

**BETA 4.8.6 | EDITION 7 NOV 2026**

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The programs are presented separately, in workflow order. Chapters explain operation, submenus and quality checks.

# Introduction and current status

This manual follows the complete workflow in the correct order: StackingWizard turns subframes into a registered linear stack; AstroWizard then handles background, colour, sharpening, denoise, stretch, finishing and export.

Verified versions: AstroWizard build 2026-11-07 and StackingWizard Beta 4.8.6. The official Linux builds were inspected directly.

- The programs are kept strictly separate; AstroWizard starts only after the Handoff.
- Automation is advice, not acceptance. Inspect the manifest, rejected frames, registration, gradients, stars and exports.
- This edition covers the new language, working-folder, background-extraction, signature, Scrying Orb, grading and memory controls.

> For later builds, read the official changelog first; labels and defaults may change during beta.

# Installation, updates and first launch

Both applications are portable. A new build replaces the program file while raw data and project results remain separate. Download only from the official AstroWizard site.

- Windows: launch the EXE; a young signature may still trigger SmartScreen.
- macOS: open the unpacked app and verify the source if Gatekeeper asks.
- Linux: unpack the archive and mark the executable as runnable.
- After a major update, process a copy of a familiar project first.

> Keep the last known-good build until the project is finished.

# File types, linear data and folders

StackingWizard mainly expects FITS subframes. AstroWizard accepts finished FITS, TIFF and XISF stacks. A 16-bit TIFF is suitable if it has not already been stretched or clipped.

- Separate Lights by target and night; preserve originals.
- Label Darks by exposure, gain and temperature.
- Capture Flats per filter and unchanged optical train.
- Archive the linear master, pre-stretch master, final, starless/stars and receipts separately.

| Folder | Contents | Check |
|---|---|---|
| Lights | Subframes | target, filter, exposure |
| Darks | Thermal signal | time/gain/temperature |
| Flats | Dust/vignetting | filter/rotation/focus |
| Results | Stacks and variants | bit depth and names |

JPEG is an output format, not a suitable linear master.

## Complete workflow and safe checkpoints

The recommended path is linear and reproducible. Save a variant before each irreversible transition.

- 1. Inspect raw data and load StackingWizard.
- 2. Grade in Looking Glass and build the stack.
- 3. Use Handoff and archive the linear master.
- 4. Process AstroWizard steps 1–6 while linear.
- 5. Step 7 stretches the image; it is nonlinear afterwards.
- 6. Finish in step 8 and document exports in step 9.

| Checkpoint | When |
|---|---|
| Linear master | immediately after StackingWizard |
| Pre-stretch | after AstroWizard step 6 |
| Nonlinear master | after step 7 |
| Final master | before web export |

Never load a finished stack together with its own subframes as Lights.

# Part I – StackingWizard Beta 4.8.6

The interface is organised as The Pile, The Looking Glass, The Stack and The Handoff. UI Scale, Workbench, Support and Fellowship sit above the workflow.

- The Pile: read the night folder and group frames.
- Looking Glass: grade and visually inspect every subframe.
- The Stack: calibrate, register, vote and integrate.
- The Handoff: save, combine and pass finished stacks to AstroWizard.

> Build 4.8.6 supports 19 languages; first launch follows the operating system where possible.

# 1 – The Pile: bring the whole night

Select the night folder itself, not individual files in the folder dialog. StackingWizard reads headers, names and folders and builds piles for Lights, Darks, Flats, Bias/dark-flats and finished stacks.

- Check target, camera, filter, exposure and frame type.
- Correct or remove misclassified files.
- Use loose files only for a deliberately limited pile.
- Load finished stacks through “Already got finished stacks?”, never as subs.

> When headers are missing or contradictory, StackingWizard votes from names and folders; clear naming improves the result.

# The Pile - import, classification and mosaic

The Pile → choose the night folder or loose files → inspect the tiles.

| Menu / control | Operation and use |
|---|---|
| Lights / Darks / Flats / Bias / Dark flats | Click a tile and select the matching folder; subfolders are included. Check headers and counts. Override a contradictory header only after checking the files yourself. |
| Unknown / Treat all as lights | Prefer assigning unknown files to the correct tile. Use the bulk option only when all unknowns really are lights. |
| Finished stacks | Already integrated masters belong here; do not add them as raw Lights. |
| Mosaic: Auto / One canvas / Panels / Off | Auto detects pointings and named panels. One canvas suits overlapping walks, Panels suits discrete mosaic fields, Off suits fixed fields. If Auto asks, use the actual capture structure rather than the target name. |
| Colour pattern: change my pick | Compare alternative readings when colours are wrong. Change CFA interpretation only for undebayered colour data; do not treat mono as Bayer. |
| Looks right / start fresh | Proceed to grading or combining only after checking the manifest. Start fresh begins a new night. |

# 2 – The Looking Glass: grade and inspect

The view shows each subframe, its grade and the machine reason. Since 4.8.4 there are two columns with a draggable divider and a bird’s-eye orientation; the whole-frame view may occupy one third, one half or be hidden.

- Play the sequence and look for jumps in focus, sky, star shape and brightness.
- Start with Standard; Ruthless is stricter and Lenient milder.
- Cloud/haze verdict does not replace checks for branches, dew, aircraft or satellites.
- Use manual grading when the automatic threshold does not suit the target.

> 4.8.6 safety net: if Standard/Ruthless would keep under 20% while the next milder grade keeps at least 75%, the milder threshold is used and explained.

# Looking Glass - manual grading limits

The Looking Glass → Grade the pile → Your lines → enter values → judge by my lines. Read values in the table columns' own units. Leave a field blank to ignore that measure.

| Menu / control | Operation and use |
|---|---|
| Sky glow | Upper limit for the displayed sky measure (%). Tight limits may reject useful exposures during moonrise. |
| Noise | Maximum measured noise. Set the limit from good and clearly poor frames rather than copying another project's numbers. |
| Stars | Minimum detected star count. Do not blindly compare sparse mosaic fields with dense fields. |
| Star size | Maximum displayed star size. Larger values can indicate focus drift or poor seeing. |
| Trailing | Upper limit for trailing. Verify elongated stars in the Magnifier. |
| judge by my lines / back to the wizard's lines | Manual limits replace automatic judgement for this night. Verdict shows failed limits; use the return control for automatic grading. No limits means no rejection by these rules, not that unreadable files become valid. |

# Looking Glass - inspect, compare and select

After grading in Looking Glass, visually review suspect frames before finalising selection.

| Menu / control | Operation and use |
|---|---|
| Magnifier / the pixel peeper / the whole frame | Click a filename, inspect enlarged stars, and use the whole frame for clouds, branches and borders. Adjust the divider and use the overview locator. |
| Set as reference / Blink vs ref | Set a good frame as reference. Blinking reveals changes in stars, brightness or artefacts. Compare similar pointings and exposures. |
| The night's chart | Open the chart and look for jumps in the measurements. Confirm each outlier in the image and verdict. |
| Keep / tick all / untick all | Toggle one frame. Bulk actions change many selections; check remaining count and coverage before Begin stacking. |
| Save the grades (CSV) / move the unticked aside | CSV records grades. Move relocates unticked files; it is not just a display option. To preserve original locations, change selection only and export CSV. |

# 3 – The Stack: build the night

The left column contains working drive, Workbench summary, unlinked view, start and receipts. The incoming frame appears in the centre and the growing stack on the right. Column dividers are draggable.

- Before “Begin the Stack”, check free space and working drive.
- Enable unlinked view when one colour channel dominates the preview.
- Watch the log, frame preview and growing stack together.
- Copy the receipts and archive them with the project.

> The working folder is never compressed. Write failures are reported per drive; use a local drive when errors occur.

# The Stack - working drive, Engine Room and retry

The Stack → verify the working drive → Begin the Stack. The two previews and the log show different stages of the same run.

| Menu / control | Operation and use |
|---|---|
| Working files / drive check | Choose a fast, local, writable drive with headroom. Use Stack anyway only after checking the warning and confirming enough space. |
| The Engine Room | Watch resource use and progress during stacking. Low free memory or a saturated drive is a reason to lower Workbench resource use on the next attempt. |
| Choose a box before the stack | Enable in Workbench before starting. On supported walk, mosaic or drizzle runs, wait for the sketch and mark the region before the final vote. This choice is for the next run only. |
| Stop / stack again / copy the receipts | Stop ends the current attempt. Stack again reruns with the current selections. Save receipts and log before clearing the night. |

# 4 – The Handoff: save and combine

The Handoff saves the finished stack, opens it in the newest AstroWizard build found, or combines multiple nights. Stacks from different telescopes or image scales are not blindly mixed.

- Keep each night’s linear stack separately.
- Use “Add other nights’ stacks” for the same target and compatible optics.
- Combine filter families separately; drizzled filter stacks are supported.
- After combining, verify new receipts and integration time.

> The Handoff button targets the newest AstroWizard, not an arbitrary older copy.

# The Handoff - save masters and combine nights

Open The Handoff after a successful stack or by selecting finished stacks in The Pile.

| Menu / control | Operation and use |
|---|---|
| Save the stack / Save the combined stack / Save all stacks | Save the intended master or save all results separately. For multiple filters, verify the filter and integration time in the filename or header. |
| Add other nights' stacks → Combine | Add at least two compatible masters of the same target. Check framing, optics and filter. Do not double-count subs through both a nightly and a combined master. |
| Same colour, different lengths | Same-colour exposure groups can stay separate or be combined with the offered control, weighted by noise. Do not mix filters unintentionally. |
| Send to AstroWizard | Save a linear master first, then process the intended stack in AstroWizard. With multiple results, do not automatically take the first. |

# Workbench – principle and scope

Every row contains the wizard’s choice and alternatives. “This session only” lasts until close; “Forever” stores decisions for future launches. “Help and examples” reveals reasons and typical cases.

- Test defaults first with familiar data.
- Change only one problem-related control at a time.
- Receipts record every departure from the wizard’s choice.
- “Reset all to the wizard’s defaults” restores the baseline.

> Workbench choices alter processing, not merely the preview.

## Workbench – quality, sky and trails

| Control | Options | Use |
|---|---|---|
| Cloud and haze verdict | Sky only / Every cell / Off | Sky only is default; Every cell for clouds across the target; Off for patchy false verdicts. |
| Outlier rejection | Normal / Gentle / Strict | Strict for surviving trails/hot pixels; Gentle for few frames or fragile signal. |
| Plane and satellite trails | Left out / Left to vote | Exclude detected trail pixels only; Vote for comparison. |
| Each frame’s own sky shape | Wizard / On / Off | Remove tilt/bowl before merging; Off for good flats and a normal fixed field. |

> Star spikes and galaxy edges remain because they repeat at the same sky position.

# Workbench – gradients, vignetting, panels

| Control | Options | Use |
|---|---|---|
| Canvas large gradient | Left in / Taken off | Left in for AstroWizard; Taken off only when no later background processing is planned. |
| Vignetting measured from the walk | Off / On | On for a Seestar/Dwarf walk without flats; Off with real flats. |
| Sample each pixel twice | On / Off | On for panel mosaics to reduce square patterns; costs about 4% star sharpness. |
| Drizzle | Off / 2x | 2x for undersampled stars with enough drift/frames; four times the pixels. |

Adaptive drizzle drop and combining drizzled filter stacks are improved in 4.8.x.

## Workbench – box, hot pixels and resources

| Control | Options | Use |
|---|---|---|
| Choose a box before stack | No / Yes | Yes only for large walks when voting would waste time; applies to the next stack. |
| Hot pixels without darks | Repaired / Left as they are | Repair uses moving stars as reference; with darks, dark calibration takes precedence. |
| PC share of final vote | Careful / Normal / Everything | Careful for ≤8 GB or slow drive; Everything only for a dedicated machine. |
| Language | 19 languages | Change takes effect at next launch. |

Deep-night streaming votes in sections after roughly 800 frames to keep RAM and working files manageable.

# Workbench - interactions and comparison runs

Workbench → This session only. For comparisons change one control, keep the same frame selection and save both receipts.

| Menu / control | Operation and use |
|---|---|
| Wolkenurteil / Ausreißer / Trails | Cloud verdict checks patches, outlier rejection compares pixel values, and the trail option masks detected streaks. A marked trail is not by itself a reason to reject the full frame. |
| Sky shape / canvas gradient / walk vignetting | Per-frame sky shape, a full-canvas gradient and edge falloff are separate corrections. Skip walk vignetting with real flats and avoid flattening nebulosity with repeated gradient correction. |
| Panel sampling / Drizzle | Panel sampling is for finished-panel mosaics; drizzle reconstructs from shifted subframes. Drizzle does not recover sampling if usable shifts are too sparse. |
| Final vote: Careful / Normal / Everything | Workbench describes Careful as about a quarter of free RAM and half the usual cores; Normal as about half of free RAM; Everything as about three quarters of free RAM and all but one core. These are resource budgets, not quality settings. |

# StackingWizard – common problems

Most failures come from classification, mismatched calibration frames, insufficient storage or an unsuitable Workbench override.

- No pile: select the folder itself and check permissions and FITS headers.
- Too many rejects: read Looking Glass and the 4.8.6 safety-net message; inspect manually.
- Dark corners: verify flats/vignetting, not merely later gradient removal.
- Square pattern in panel mosaic: test “Sample each pixel twice”.
- Stack stops: check working drive, free space, file system and Careful mode.

> Receipts and log belong with every bug report; they document the choices actually used.

# Part II – AstroWizard build 2026-11-07

AstroWizard guides you through nine steps. The order protects linear tools from an early stretch and keeps Undo, Compare and saved checkpoints understandable.

- 1 Load & Inspect
- 2 Crop & Framing
- 3 Remove Gradient
- 4 Balance Colour
- 5 Sharpen · 6 Reduce Noise · 7 Stretch · 8 Final Polish · 9 Export

> Leave a step only when the main image, 100% view and Compare all look plausible.

## Settings & Help – complete overview

The settings dialog combines language, engines, display, online services and working files. Selections are stored.

- Language: one of 19 languages.
- Assign GraXpert, StarNet2 and RC-Astro CLI with “Locate”; RC-Astro CLI can be updated.
- Enable online version/Fellowship and Gaia access as needed.
- “Hide free tools where RC-Astro does job” simplifies the interface.
- Set UI Scale and Live Console for the display and diagnostic need.
- Working files: beside the picture or in a chosen folder.

> In a chosen working folder each image gets a subfolder; it is cleared on close, start-over or loading another image.

# Settings & Help - engines, models and privacy

Settings & Help → AI Engine Configurations / AI Compute & Models. Check executable paths, compute device and model version separately.

| Menu / control | Operation and use |
|---|---|
| Locate / Update RC-Astro CLI | Select GraXpert, StarNet2 and one shared rc-astro executable for BXT/NXT/SXT. On Windows the RC-Astro CLI folder is a typical location. Verify the path after updating if an engine disappears. |
| RC-Astro Licences | Shows tool status. An activation code from your RC-Astro account covers the tools you own. Do not include codes in screenshots or bug reports. |
| Auto / GPU / CPU / Which GPU | Start on Auto. Try CPU for diagnosis if the GPU fails, or choose a specific device on multi-GPU systems. Repeat the same small test after changing settings. |
| Model version per tool | Latest uses the engine default. BXT, NXT, SXT, GraXpert BGE and GraXpert denoise are independent. Specify a version only if the installed engine supports it. |
| Updates & Privacy / Hide the free tools | The online switch covers version checks, Fellowship and Gaia. With free alternatives hidden and RC-Astro located, GraXpert background extraction remains available. |
| UI Scale / console / Working files | Scale the interface and restart on Linux if needed. The console shows engine messages. Do not use temporary working files as a project archive. |

# Step 1 – Load & Inspect

Load the linear master from StackingWizard. AstroWizard reads format, dimensions, channels, header, WCS, integration time and existing processing hints.

- Prefer FITS/XISF; use 16-bit TIFF only while unstretched.
- Inspect black/white points and channel statistics for clipping.
- Note mosaic borders, registration wedges, amp glow and flat errors.
- For already stretched data, acknowledge the warning deliberately.

> Start over reloads the initial state; a new image discards the current temporary history.

# Step 1 - loading paths and channel mixing

Load Your Image: the combiner is in step 1, not Final Polish. It operates on input stacks, not finished JPEGs.

| Menu / control | Operation and use |
|---|---|
| Load Normal Image (RGB/OSC) | Load a finished colour stack. Try unlinked display for a strongly tinted preview; actual stretch values are set in step 7. |
| Combine Mono Channels | Load registered mono channels individually with Browse. Broadband maps R/G/B, SHO maps S/H/O, and HOO places H in red and O in green and blue. Luminance is optional for suitable broadband data. |
| My own mix | Set To red / To green / To blue to your intended mapping. Do not infer filters from preview colour; verify headers, filenames and registration. |
| Two dual-band stacks → SHO | Assign Ha/OIII and SII/OIII correctly. The app registers the stacks by their stars. Check overlap and the S/H/O mapping afterwards. |
| Load Planetary Data / Library / Guide You | Planetary mode is a separate short route (Framing, Sharpen, Polish, Export). Library offers practice data. Profile guidance highlights recommendations but does not replace inspection. |

# Step 2 – Crop & Framing

Cropping removes unreliable stack borders and establishes composition. WCS is updated. Geometric changes also remain synchronised across existing star/starless layers.

- Artifact Crop: 2.5 / 5 / 7.5 / 10% for symmetric edge cleanup.
- Free Crop for unconstrained composition.
- Ratios: 1:1, 16:9, 4:5, 3:2, 4:3, 2:1 and 9:16.
- Flip Horizontal/Vertical and Rotate CW/CCW.
- Use Undo before setting a new crop.

> Remove real artefacts only; an overly tight crop limits later variants and mosaic joins.

# Step 2 - free crop and rotation

Crop & Framing → Free Crop. Drag a box, resize by its corners, or drag inside to move it.

| Menu / control | Operation and use |
|---|---|
| Lock / Tilt ° / Set | Lock preserves the selected aspect ratio. Enter a Tilt angle and use Set or Enter; dragging just outside the box rotates the image behind the crop. Enter applies the selection; Esc cancels. |
| Orient (Optics) / Apply Angle | Flip or rotate in 90° steps; for a small correction use the fine rotation dial or angle and choose Apply Angle. Inspect preview and final borders. |
| Bin 2×2 | Halves width and height. Can reduce memory and noise per resulting pixel, but costs spatial resolution. For a drizzled stack consider the additional option after sharpening. |
| Reset Step / Last-minute crop | Reset Step discards edits for this step. Use Last-minute crop for late export framing, then save and verify dimensions. |

# Step 3 – built-in Background Extraction

The new internal extraction offers Auto and Manual. “Light pollution” subtracts an additive gradient; “Vignetting” divides a multiplicative model. Do not confuse them.

- Auto: generate a suggestion, then inspect model and result.
- Manual: place representative sky samples and exclude target areas.
- “Hold gradient taken” displays the removed model.
- “Hold sky it read” displays what was interpreted as sky.
- Show Tips reveals contextual guidance.

A good model contains large-scale illumination but no nebula arms, galaxy halo or dust structure.

## Step 3 – Manual: marks and dials

The manual view separates marking tools and model controls. Clicking a star samples nearby sky; orange marks indicate outliers.

| Control | Function |
|---|---|
| Suggest | Place automatic starting samples. |
| Spot | Place one sky sample. |
| Box | Mark an area as a sample. |
| Not sky | Exclude target/nebulosity from sky measurement. |
| Undo mark / Clear | Remove last mark / clear all marks. |
| What is sky | Sky-detection tolerance; starting value about 0.80. |
| Gradient scale | Fine, Normal, Broad, Broadest – scale of modelled structure. |
| Pip size | Sample size, commonly 32 px on auto. |

Fine may model real target structure; Broadest may miss local gradients. Let the model view decide.

# Step 3 – GraXpert and decision

GraXpert remains available below the internal extraction. Background Smoothing offers Default, Less, More and Custom; “Hold view extracted gradient” displays the model.

- Compare the built-in extraction first on a familiar image.
- For complex IFN/nebulosity, save and compare both models.
- Reset Step resets step 3 only.
- Never stack two strong gradient corrections.

> Vignetting from bad flats belongs in calibration; gradient tools can only conceal it visually.

# Steps 3 and 4 - inspect the model and use Gaia

Apply background and colour tools while linear. A before/after comparison is more informative than an aggressive screen stretch.

| Menu / control | Operation and use |
|---|---|
| Manual: Marken bewegen und entfernen | Use Suggest as a starting point. Drag a mark to move it or right-click to remove it. Exclude nebulosity, galaxy halos and dust with Not sky. Assess dials and marks separately. |
| Custom GraXpert Extraction | Background Smoothing → Custom → Smoothing Factor (0–1; opens at 0.50) → Run GraXpert. Change one setting at a time and use Hold: View Extracted Gradient to check whether target structure entered the model. |
| Gaia-Status / True colours from the stars | When Gaia data are ready, click to match. After a network failure, the click retries retrieval. If sky coordinates are missing, load a suitable linear FITS stack with coordinates. |
| Background Neutralization | Neutralise background when the cast is in the sky. For narrowband, do not treat it as a substitute for the intended HOO/SHO mapping. Avoid double correction when another colour route already looks right. |

# Step 4 – Balance Colour

Background Neutralization removes residual neutral cast. “True colours from the stars” uses stellar measurements and Gaia when linear broadband data, WCS/coordinates and enough clean stars are available.

- Run Background Neutralization first and inspect the background.
- Use Star-True Colours only on linear broadband data.
- Verify the plate-solved target and crop frame.
- Do not force dual/narrowband data into supposed natural colour photometrically.

> True Colours corrects calibration, not clipped or previously desaturated stars.

## Step 4 – colour diagnosis and magenta stars

Judge channel statistics and star colours together. “Correct Magenta Stars” preserves brightness better in current builds; still inspect rings and pale cores at 100%.

- Green background: check neutralisation/gradient before reducing saturation.
- White stars: clipping usually occurred earlier; return to pre-stretch.
- Apply magenta correction only to truly affected stars.
- Hold Compare Original and judge background, target and stars separately.

> The screen preview is sharper; file values change only on Apply/commit.

# Step 5 – Sharpen

Choose Native High-Pass or optional BlurXTerminator, each with Light, Normal, Heavy and Custom. Sharpening should separate structure, not shrink stars at any cost.

- Start with Light on already good data.
- Use Normal only with stable stars and sufficient signal.
- Heavy is rare; inspect bright-star halos and dark rings.
- Engine buttons are protected against accidental double-clicks.

> With poor seeing, stronger sharpening usually creates artefacts rather than real detail.

## Step 5 – Custom High-Pass

The Custom dialog separates amount, radius and Halo Guard. Preset neighbourhoods are approximately Light 0.20, Normal 0.40 and Heavy 0.70; default radius is about 1.5 pixels.

- Amount controls strength.
- Radius selects feature size; small values emphasise fine detail and noise.
- Halo Guard limits rings around bright stars and hard edges.
- Inspect at 100% and Fit; the views reveal different problems.

> Several small applications are easier to control than one extreme pass.

# Step 5 - Custom BlurXTerminator

Sharpen → BlurXTerminator → Custom → Run BXT. Fields are from the documented build; a licensed processing run was not tested.

| Menu / control | Operation and use |
|---|---|
| Sharpen Stars | Controls stellar sharpening. Reduce it first if cores become hard, stars look unnaturally small, or rings appear. |
| Star Halos | Negative shrinks halos, positive grows them. Do not use strong shrinkage to conceal excessive stellar sharpening. |
| Sharpen Nonstellar | Sharpens nebula and galaxy structure. Back off if background grain sharpens or filament-like artefacts appear. |
| Automatic PSF / PSF Diameter (px) | Start with automatic PSF. If switched off, use typical stellar width in pixels as a starting estimate; tight, crowded fields may require a smaller estimate. |
| Planetary Mode / drizzled stack | Planetary Mode shows nine previews for lunar/planetary data. On deep-sky drizzled stacks, optional Bin 2×2 after sharpening can return to native sampling. |

# Step 6 – Reduce Noise

GraXpert AI and NoiseXTerminator offer Light, Normal and Heavy; NoiseX also offers Custom. The goal is a quieter background while preserving fine structure.

- Denoise after background and sharpening, but before stretch.
- Begin with Light and inspect the faintest target region at 100%.
- Waxy surfaces, smeared small stars and colour blotches indicate overprocessing.
- Do not try to erase noise caused by insufficient integration.

> The best denoise looks calm at Fit and retains natural fine grain at 100%.

# Step 6 - Custom NoiseXTerminator

Reduce Noise → NoiseXTerminator → Custom → Run NXT. Compare one pass before adding another. Control labels checked; no licensed processing run was tested.

| Menu / control | Operation and use |
|---|---|
| Denoise strength (overall) | Overall strength. Start low enough to retain faint edges and small stars naturally. |
| Intensity (luminance) denoise | Targets brightness noise. Too much gives waxy areas or thins fine detail. |
| Colour (chrominance) denoise | Targets colour noise. Too much flattens faint real colour; check both low-colour areas and nebulosity. |
| Fine grain / Coarse blotch | Weight small grain and large blotches separately. Check larger blotches for residual gradients first; denoise is not gradient repair. |
| Scale boundary (px) | Sets the fine/coarse boundary in pixels. Compare the same region at the same zoom. |
| Iterations (NoiseXTerminator's own) | Engine-internal iterations, not repeated AstroWizard passes. Avoid raising all strengths and iterations together. |

# Step 7 – Stretch Wizard

The Stretch Wizard develops linear values into a visible nonlinear image. “Keep this look” retains the current preview; a banner indicates the held state.

- Choose the target profile and inspect the preview.
- “Image looks all one colour?” enables unlinked stretch for channel imbalance.
- Watch star cores, background and faint signal together.
- Save the pre-stretch master before committing.

> If stars turn white or shadows clip, go back; step 8 cannot restore lost values.

## Step 7 – Manual GHS and Legacy

“Open the dial” opens Generalized Hyperbolic Stretch. Histogram and transfer curve show symmetry point, shadow/highlight protection and strength. The dialog now applies each pass once; “Reset and start again” restores the start.

- Place Symmetry Point on the relevant signal range.
- Shadow Protection prevents black clipping.
- Highlight Protection preserves star cores.
- Use small passes, then inspect the main image and Compare.
- Legacy: Classic Normal/Aggressive/Custom and Deep/Deep keep colours.

> Manual passes are not silently repeated; each click is one new pass.

# Step 7 - Stretch Wizard, GHS and Legacy in detail

Stretch Wizard: choose colours, then depth. Scroll to zoom all panes, drag to pan, click to apply a preview, or Esc to cancel. The star symbol marks the suggestion.

| Menu / control | Operation and use |
|---|---|
| Transformation: Generalised Hyperbolic / Linear / Invert | GHS shapes the tonal curve; Linear uses Black Point and White Point. Invert reverses the transformation, not the image display colours. |
| ln(D+1) / Local intensity (b) | The stretch factor sets strength. b shapes concentration around the symmetry point. Observe changes in histogram, transfer curve and preview. |
| SP / Pick SP from the picture / SP = the sky | SP sets the region of strongest effect. Pick a picture point or the sky as a starting point and compare small changes. |
| LP / HP / BP / WP | LP/HP protect shadows/highlights in GHS. BP/WP set black/white points in Linear mode. Check the histogram for clipping. |
| Apply / Hold to compare / Reset dial / Reset and start again | Apply builds on the current image. Hold to compare checks the pass or step entry, depending on state. Reset dial resets controls; Reset and start again restores the input to step 7. |
| Legacy Custom: Classic / Deep (anchored) | Classic sets factor (Normal 150, Aggressive 600 for reference). Deep targets sky brightness (Deep about 13%). Deep (keep colours) better retains channel balance. Check preview before applying. |

# Step 8 – Curves and Masked Saturation

Curves controls luminance, RGB channels, saturation and colour families. The fast preview is followed by a full-resolution calculation after a short pause.

- RGB/luminance: small S-curve with background and highlight anchors.
- R/G/B: correct colour cast by brightness range.
- Saturation: lift weak colour while protecting strong colour.
- Colour family: target blue/cyan, red and so on.
- Dose Masked Saturation with Custom Effect Strength; 0.50× is the dialog's one-click reference.

> Saturation cannot create missing OIII; if only chroma noise grows, narrow the mask or back off.

## Step 8 – Narrowband Tune and Colour Spells

Narrowband Tune organises Ha, OIII and SII. Colour Spells are stackable, editable starting recipes, not mandatory presets.

| Tool | Effect | Check |
|---|---|---|
| Yellow to Copper & Gold | Yellow → copper/gold | SHO, stars and background |
| Blue to Teal | Blue → OIII teal | Do not cyan-shift the whole sky |
| Dynamic Pop | gentle finishing S-curve | reduce after strong contrast |
| Add Warmth / Cool Down | warmer/cooler overall | watch star colours |

> The book icon opens the recipe in Curves. For SHO: Narrowband Tune first, at most one spell, then small manual curves.

# Step 8 - cards, palettes and masks

Final Polish expects stretched data. Cards group related tools; palette visibility depends on the input profile.

| Menu / control | Operation and use |
|---|---|
| Foundation & Palettes | OSC Dual-Band to Hubble is for compatible colour-camera dual-band data. MONO SHO to Hubble is for separate S/H/O channels. Use Auto Colour Balance after good Star-True calibration only if needed. |
| Background & Contrast | Darker Sky / Lighter Sky alter the background presentation. Check faint signal and black point after each pass; these are not linear gradient correction. |
| Colour & Enhancements | Masked Saturation targets the mask; Make it Pop increases punch. Custom Effect Strength offers Apply Saturation / Apply Pop; its one-click reference is 0.50×. Repeated Apply stacks. |
| Utilities: Hold View Mask / Mask Reach | Hold to inspect the mask. A lower Mask Reach includes fainter structures; Set accepts, Default restores 1.2, Cancel discards. Back off if it reaches chroma noise. |
| Green-Be-Gone / Correct Magenta Stars / final NR | Use green correction selectively, not on wanted OIII green. Check magenta star correction at 100%. Use Final Noise Touch-Ups (Light/Normal, GraXpert/NXT) only as optional finishing. |

# Step 8 - Curves: editing curves

Open Curves (Colour Targeting) or the book beside a spell. Curves requires a colour image; compare preview and main image.

| Menu / control | Operation and use |
|---|---|
| Channel / Target | Choose brightness, an individual RGB channel or saturation; Target limits editing to a colour family. Check which regions actually change. |
| Punkte / Vorschau | Click adds a point, drag shapes the curve, right-click removes a point. Scroll zooms the preview, drag pans, and double-click fits it. |
| Compare / Apply | Hold: Compare Original provides the baseline. Apply commits while leaving the window open. Do not apply again if the intended effect is already achieved. |
| Reset Channel / Reset All / Close | Reset Channel affects the active curve; Reset All clears the dialog’s curves. Close dismisses the dialog; use history/Undo for committed edits. |

# Step 8 - Narrowband Tune: palette controls

Final Polish → Foundation & Palettes → gear beside the available HOO/SHO palette. The colour profile determines which variant is available.

| Menu / control | Operation and use |
|---|---|
| Synthetic green / Blend amount | Compare Mode1, Mode2 and Mode3 in preview; Blend amount adjusts the mix. The number is not a quality rating. |
| OIII boost / SII boost | Boost the relevant line component, depending on palette. In faint areas, check whether structure or only chroma noise increases. |
| SCNR (remove green) | Reduce green. Use small steps: faint OIII signal can itself appear greenish in the chosen palette. |
| Shadow point / Highlight reduction / Brightness | Adjust shadow point, highlight reduction and overall brightness separately. Check faint signal, bright stars and nebula cores together. |
| Apply / Cancel | Inspect dialog and main image; Apply commits and Cancel aborts the trial. |

# Step 8 – remove and replace stars

Remove Stars uses StarXTerminator or StarNet2. Current builds save “Stars Only” both pure and as a laid-layer variant; “Save all” carries both.

- Save a nonlinear master before separation.
- Inspect starless at 100% for holes, rings and target remnants.
- Do not alter geometry outside the synchronised workflow after separation.
- Replace Stars with a natural mix; 1.0 is full restoration.
- Inspect bright, medium and small stars separately.

> Star separation is optional. A harmonious global treatment is often better than an artificial starless look.

## Cross-reference to step 1 - Combine and “My own mix”

The combiner preserves header information and offers “My own mix” for custom channel assignments. Dual-band combinations can provide HOO/SHO starting points; registration and image scale must be compatible.

- Keep inputs linear and geometrically identical.
- Verify channel roles and filter labels before commit.
- Check integration metadata in the combined header.
- Record custom mixes or save the receipt.

> An attractive preview mix does not replace correct physical filter mapping.

# Step 8 - Replace Stars: brightness, colour, blend

Final Polish → Star Management → Remove Stars (SXT or StarNet2) → Replace Stars. Save a master before separation and inspect starless residuals.

| Menu / control | Operation and use |
|---|---|
| Star Brightness | 1.00 is the reference for exact restoration in additive mode. Start below it to reduce dominance, then check small stars and halos. |
| Star Colour | 1.00 is the natural reference in the dialog. Higher values deepen gold/blue tones before blending; tune separately from Brightness. |
| Exact (additive) / Soft (screen) | Exact adds the separated component. Soft uses a screen blend and is gentler over bright areas. Do not expect identical star rendering at equal slider values. |
| Live preview / Cancel | The main canvas previews the result; use zoom and pan to compare. Commit only when bright and faint stars look right. Cancel discards the trial. |

# Step 9 – Export & Share

Export is part of quality control. Save several clearly named levels, not only a small JPEG.

| File | Recommendation | Purpose |
|---|---|---|
| Linear master | FITS/XISF, full precision | complete reprocessing |
| Pre-stretch | after step 6 | new stretch |
| Final master | 16-bit TIFF or FITS/XISF | archive/print |
| Starless / Stars | high bit depth | later variants |
| JPEG/PNG | sRGB, target size | web/sharing |

FITS is not signed; labelled output is saved as JPEG, PNG or TIFF and as a separate labelled copy.

## Create a signature

The Signature Creator places words or a logo, offers six built-in fonts or a custom font file, plus position, offset, opacity and an optional integration-time line.

- Place the signature small and away from important image structure.
- Check opacity against both dark and bright background.
- Optionally include it in Scrying Orb.
- Always preserve an unsigned master separately.

A signature is not written to FITS, starless or stars-only files.

# Step 9 - Save all, signature and labelled copy

Export & Share → Save Final Image or Save all project files. Starless, Stars Only and Stars as laid appear according to the edit state.

| Menu / control | Operation and use |
|---|---|
| Into / Change / Prefix / Formats | Set destination and filename prefix. Select JPG, TIFF and FITS individually in the batch dialog. Name collisions are numbered. Check the actual files before sharing. |
| Starless / Stars Only / Stars as laid | Starless = target without stars. Stars Only = separated star component. As laid = stars as adjusted for brightness/colour in Replace. Without replacement, the final may already be identical to Starless. |
| My signature: Words / Picture / Font / Size | Choose text, a logo or both. Choose loads a logo, Remove clears it from the signature. Font offers built-in options or My own font file. Check size in the preview. |
| Place / Edge / Look / Solid logo | Check horizontal/vertical placement, edge margin and appearance. Solid logo can firm up a faint graphic. Test bright and dark image edges. |
| Integration / unsigned copy / Sign my finished pictures | The optional extra line shows integration data when present. Unsigned copy keeps a clean variant; the signed copy gets _signed. FITS, starless and star layers are not signed. |
| Name what’s in my picture | Label the galaxies, nebulae and clusters makes a labelled copy after matching stars to the sky. Check placement and keep an unlabelled master. |

# Scrying Orb – process animation

Scrying Orb creates GIF, MP4 or both from saved processing states. Current builds retain every step and provide more reliable undo/redo.

- Choose GIF, MP4 or Both.
- Include the signature if desired.
- Watch the full animation before publishing.

> A long history is valuable documentation, but a short selection is clearer to viewers.

# Step 9 - Scrying Orb: export dialog

Export & Share → Replay my edit as a GIF or a video. The current session needs at least one edit.

| Menu / control | Operation and use |
|---|---|
| Save as | Choose GIF for a looping preview, MP4 for video, or Both. Open the actual output before sharing. |
| Width (px) / Seconds per step | Choose 720, 1080 or 1440 px wide, then 1, 2 or 3 seconds per step. Test a small preview first for long histories. |
| Name each step / Processed with AstroWizard | Step captions and program credit are separate toggles. Keep step names for a learning version; inspect visually for presentation use. |
| Sign it with my signature / Create the orb | Set up My signature first, then enable it here separately. Create the orb renders the recorded edit history. Play through the full sequence before publishing. |

# AstroWizard – common problems and recovery

Undo, Compare, Reset Step and Start over have different scopes. Use the smallest suitable recovery so good earlier work is preserved.

- Gradient contains target: correct marks/scale or compare GraXpert.
- Image looks one colour after stretch: test unlinked view/stretch and channel balance.
- Dark star rings: reduce sharpening/starless processing.
- Waxy background: reduce denoise.
- White stars/black sky: return to step 7 and protect highlights/shadows.
- Missing engine: check path, executable bit, weights/licence and Live Console.

> The main display now follows history rewind; still compare saved masters outside the app.

## Quick recipes by data type

| Data | Recommended path |
|---|---|
| Broadband galaxy | Gradient → Neutralize → True Colours → light sharpen/denoise → protected stretch → small luminance/saturation curve. |
| Dual-band HOO | Gradient → no false photometry → denoise → colour-preserving stretch → Narrowband Tune → selective cyan. |
| SHO | Verify channel roles → stretch → Narrowband Tune → one spell → Curves → inspect stars. |
| IFN / faint dust | Very cautious Background Extraction → low sharpening → moderate denoise → many small GHS passes. |

> Recipes are starting points. Target, integration, sky and optics determine the actual strength.

# Final checklist

Before final export, inspect the image at Fit, 100% and in a second viewer.

- No clipped shadows or blown star cores.
- Background even, but not sterile or black.
- Stars round, coloured and free of dark rings.
- Faint signal visible without chroma noise dominating.
- Linear, pre-stretch and final masters present.
- Receipts, build numbers, filter mix and special Workbench values documented.
- Web file in sRGB, correct size and without accidental metadata loss.

> The image is finished when another change brings no clear gain without new damage.

## Version notes and sources

This edition was prepared for AstroWizard build 2026-11-07 and StackingWizard Beta 4.8.6. It is based on the official program interfaces, official changelog and the previous edition dated 18 September 2026.

Menu labels refer to the English interface of the documented builds. Profile-, file- and licence-dependent features appear only when their prerequisites are met. The reference describes checked dialogs; it is not evidence of an end-to-end test of every variant.

- Official home: https://astrowizard.lukomatico.com
- Changelog: via the official AstroWizard page.
- StackingWizard: via the official StackingWizard link.
- Gaia stellar data: ESA Gaia, through services used by AstroWizard.

> Beta notice: features may change. Preserve masters and receipts independently of the application.